

Consult 360 AI

Next-Generation Clinical Decision-Support & Longitudinal Care Journey Copilot

Project Repository:	github.com/divyansh24bce10143-coder/CONSULT-360-AI
Core AI Engine:	Google Gemini 3.6 Flash (Multimodal Vision & Structured Synthesis)
Target Audience:	Hospital Outpatient Departments (Cardiology, Endocrinology, Internal Medicine)
Architecture:	Node.js / Express backend · Vercel Serverless · Real-time SSE streaming · Zero data retention
Release Version:	v2.4 Production Release · Clinical Design System

1. Executive Summary & Clinical Rationale

Consult 360 AI is a high-performance clinical decision-support copilot designed specifically for physicians in busy outpatient and ambulatory clinic environments. In contemporary healthcare systems, clinicians are allotted merely **5 to 10 minutes per patient consultation**, while being required to manually digest over 20+ pages of fragmented medical records, PDF lab sheets, 12-lead ECG strips, 2D echocardiograms, and handwritten doctor notes.

This extreme cognitive load results in diagnostic fatigue, medication reconciliation errors, and missed preventive care interventions. Consult 360 AI solves this by acting as an intelligent pre-consultation engine: it synthesizes multi-modal patient documents in real time, computes longitudinal biomarker trajectories, maps the patient across an interactive **6-Stage Care Journey Continuum**, and presents an actionable, explainable clinical brief with human-in-the-loop decision controls.

Key Medical Impact: Consult 360 AI reduces physician chart-review time from 6–8 minutes down to under 45 seconds, while proactively catching silent clinical deteriorations (e.g. accelerating proteinuria, subclinical eGFR declines, drug-allergy conflicts, and overdue ADA/KDIGO guideline investigations).

2. Core Platform Capabilities & Clinical Features

A. Multi-Modal Medical Document Ingestion

Unlike generic LLM wrappers that only process plain text, Consult 360 AI provides a specialized vision-OCR pipeline capable of ingesting complex, multi-format medical diagnostic records simultaneously:

- **12-Lead ECG / EKG Waveforms:** Automatic extraction of Heart Rate, Rhythm (AFib/Sinus), PR interval, QRS duration, QTc, ST-segment deviations (STEMI/NSTEMI), axis shifts, and bundle branch blocks.
- **2D Echocardiograms & Doppler:** Measurement of Left Ventricular Ejection Fraction (LVEF %), regional wall motion abnormalities (RWMA), and valvular regurgitation/stenosis grading.

- **Chest Radiographs (X-Rays):** Extraction of cardiothoracic ratio, pulmonary infiltrates, pleural effusions, and consolidation patterns.
- **Handwritten Prescriptions & Clinical OPD Notes:** Transcription of handwritten medical scripts, drug strengths, dosing intervals (OD/BD/TDS), and historical diagnoses.
- **Digital & Scanned Laboratory PDFs:** Structured extraction of biochemistry, hematology, and lipid profiles with normal reference threshold evaluation.

B. The 6-Stage Care Journey Continuum

Consult 360 AI replaces static summaries with a dynamic care continuum state machine, tracking every outpatient through 6 standardized hospital stages:

Stage	Clinical Definition	Status States Tracked
1. Consultation	Initial patient intake, chief complaints, and encounter logging.	Completed / Pending
2. Diagnosis	Confirmed and suspected diagnoses with ICD-10 codification.	Confirmed / Active / Ruled Out
3. Treatment	Reconciled pharmacological therapy, dose titrations, and contraindications.	Active / Titrated / Discontinued
4. Investigation	Diagnostic labs, imaging, and pathology audits against clinical guidelines.	Completed / Requires Attention / Overdue
5. Follow-up	Longitudinal multi-visit appointments and interval monitoring schedules.	Scheduled / Overdue / Missed
6. Review	Multi-specialist chart evaluation and final attending physician sign-off.	Pending / Authenticated

C. Explainable AI (XAI) Risk Signals & Clinician Overrides

To ensure safe and accountable clinical decision support, Consult 360 AI strictly prohibits opaque 'black-box' recommendations. Every risk flag incorporates mandatory XAI fields:

- **Biological Reasoning:** Clear pathological mechanism explaining why a parameter poses acute or chronic risk.
- **Cited EHR Evidence:** Explicit excerpts and numerical biomarkers extracted verbatim from the patient's records.
- **Source Document Provenance:** Identifies the exact origin report (e.g. 'Lab_Report_March2024.pdf') and encounter date.
- **Confidence Scoring:** Calibrated probabilistic grounding score (e.g. 94% confidence).
- **Human-in-the-Loop Override Bar:** Physicians can interactively execute [✓ Accept & Order], [■ Specialist Referral], or [X Dismiss Flag].

D. Longitudinal Metric Trends & Delta (Δ) Shift Engine

The system performs automated chronological comparison across multiple patient encounters, calculating exact percentage biomarker shifts (Δ **Shift**) for critical parameters like HbA1c, eGFR, Blood Pressure, and Lipid fractions. Each parameter is tagged with clear directional clinical indicators (*Worsening*, *Improving*, or *Stable*).

E. Active Chart Records & Encounter Order Sheet

All verified physician directives, accepted lab orders, and specialist referrals are tracked live in an interactive **Encounter Chart Record**. Clinicians can view active orders in real time, export formatted summaries to hospital EHRs (Epic Systems, Oracle Cerner) via 1-click clipboard copy, or generate formal printable physician order sheets.

3. Technology Stack & Architectural Specification

The system is built on a modern, decoupled client-server architecture designed for sub-second latency, high security, and serverless scalability:

Component Layer	Technology / Framework	Architectural Role & Functionality
Generative AI Engine	Google Gemini 3.6 Flash (@google/generative-ai)	Multimodal vision OCR (ECG, Echo, X-ray, Rx), medical entity recognition (NER), guideline inference, and structured JSON synthesis.
Backend API Server	Node.js · Express.js · Multer	REST routing, in-memory buffer streaming (zero disk storage), and resilient API retry with exponential backoff.
Real-time Stream	Server-Sent Events (SSE)	Provides live 7-stage pipeline telemetry to the browser (Ingestion → OCR → NER → Timeline → Delta → Risk → Synthesis).
Deployment & Cloud	Vercel Serverless Functions	Global edge distribution via api/index.js entry point and root vercel.json rewrites.
Frontend UI Framework	Modern Vanilla HTML5 / ES6+	Zero bloated dependencies. Fast DOM rendering, responsive CSS grid, modal controllers, and asynchronous fetch handlers.
Clinical Design System	Custom Healthcare CSS3	Hospital slate palette (#f8f9fc), deep navy typography, triage badges, Care Journey stepper, and print stylesheets.

Document Processing	PDF.js · Multimodal Vision	Client-side PDF text extraction and fallback multimodal image OCR for scanned and handwritten documents.
---------------------	----------------------------	--

4. Real-Time 7-Stage Analysis Pipeline Flow

When patient records are ingested into Consult 360 AI, they pass through a sequential, fault-tolerant clinical pipeline:

1. Ingestion	Multi-file upload buffer parsed in memory (PDFs, ECGs, Echo scans, Prescriptions).
2. Multimodal OCR	Gemini 3.6 Flash transcribes handwriting, waveforms, and visual scan features.
3. Entity NER	Extraction of Vitals, Diagnoses (ICD codes), Active Medications, and Biomarkers.
4. Timeline Sync	Longitudinal chronological indexing across all historical and current encounters.
5. Delta (Δ) Engine	Quantitative mathematical comparison of biomarker shifts and trajectory slopes.
6. Clinical Risk Audit	Cross-referencing contraindications, drug interactions, and care gaps against guidelines.
7. Brief Synthesis	Generation of the pre-consultation summary, action checklist, and structured brief.

5. Security, Privacy & Compliance Highlights

- **Zero Permanent Data Retention:** Ingested files are held exclusively in volatile RAM buffers during synthesis and discarded immediately post-analysis.
- **Server-Side Credential Isolation:** API keys are isolated within server-side environment variables (.env) and are never exposed to the client browser.
- **Human-in-the-Loop Governance:** AI outputs are explicitly framed as decision-support suggestions; orders and prescriptions require attending physician sign-off.

Document Prepared By: Consult 360 AI Development Team
Repository: divyansh24bce10143-coder/CONSULT-360-AI

Status: Production Certified
Verified Engine: Gemini 3.6 Flash Active